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Key node identification method based on hybrid centrality and graph neural network

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Abstract

Key node identification has become an important part of complex network research. However, existing key node identification methods suffer from low computational efficiency and insufficient robustness. This paper proposes to obtain a mixed centrality by combining the local and global influences of nodes, and after combining it with the graph convolutional neural network, a key node identification model is proposed. The model constructs multi-scale centrality features using mixed node centrality and builds two structural channel sets to embed neighborhood topology information. Subsequently, an attention mechanism is introduced to assign weights to different channels automatically. Finally, feature aggregation and regression prediction of nodes are carried out by the graph neural network. Experimental results on multiple real social networks and synthetic scale-free networks show that NLGCN outperforms classical centrality methods and existing graph learning models in terms of propagation ability, node ranking consistency, robustness, and generalization performance.

Keywords

Social networks; key nodes identification; graph neural networks; hybrid centrality; attention mechanism

1. Introduction

The rapid development of the Internet has led to the continuous growth of the scale and complexity of social networks. As a core issue in social network analysis, key node identification is widely used in public opinion control, epidemic prevention and control, information recommendation, supply chain optimization, and other scenarios [1-3]. Its goal is to identify nodes with high influence in communication and structural control. Accurate identification of these nodes has important practical significance for improving transmission efficiency and enhancing network robustness. Practical applications in real life include controlling the spread of disease by identifying infected individuals, quickly detecting and suppressing network attacks [4-7], accelerating the response and flow of information [8], and gaining insight into high-quality suppliers in the supply chain [9-11]. The key to solving such practical applications is to measure the influence of nodes in the network, identify the Top-N nodes with the highest influence, and then use them to guide the solution of real-world problems.

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In the existing influence node identification methods, methods based on local influence, such as DC, H-index and its variants, and methods based on global influence, such as PageRank, Eigenvector Centrality (EVC), Betweenness Centrality (BC), etc [12-15]. It performs well in some scenarios, but these methods have the following limitations: algorithms based on local influence will lose global information, which may not be ignored in some networks; the algorithm based on global structure will have problems such as complex calculation and high time redundancy when encountering large, complex networks [16-20]. These limitations will lead to problems such as insufficient expression ability and poor generalization of the algorithm in the face of large-scale networks with complex multi-scale structures, which cannot deal with high-order or dynamic structures [21-24]. To solve this problem, some scholars use a graph neural network (GNN) to automatically learn the potential influence of nodes to improve the generalization ability and prediction performance of the model. For example, GCN and GraphSAGE perform well in node classification and link prediction tasks. The GNN model supports modeling with complex structures, and its generalization ability has been improved. However, model training is sensitive to network topology. If there are nodes/edges missing, the model performance may be greatly reduced, and its robustness cannot be guaranteed.

Considering the above problems in key node identification in social networks, this paper proposes a hybrid centrality feature (i.e., the local and global influence of hybrid nodes) and a graph convolutional neural network key node identification model, NLGCN. The main contributions are as follows:

1. To take into account the local information and global information of nodes and improve the computational efficiency at the same time, this method chooses to use the mixed centrality method, that is, to fuse the local influence (NLI) and global influence (NGI) of nodes, and introduce the K-order neighbor sum into the representation of the local influence of nodes, to further optimize the representation of the local structural characteristics of nodes, and use the shortest path and node degree to calculate the global influence, replacing the K-shell value to improve the computational efficiency.
2. To improve the robustness of NLGCN, the self-attention mechanism is introduced to adjust the channel weight, which improves the adaptability of the model under input changes, no longer relies on the fixed structure path, and improves the robustness of the model under neighborhood disturbance.
3. To avoid sacrificing computational efficiency while improving robustness, this method uses a one-hop neighborhood to construct an effective channel set, which greatly reduces the computational complexity. Six channels are constructed by multi-scale centrality, which are divided into two channel sets: local influence and global influence. The unified constructed intra-channel features make the features share the input structure when the model propagates forward, avoiding redundant convolution.

The rest of this paper is organized as follows: In Section 2, we briefly review the related work. Section 3 introduces the framework of the proposed method. Section 4 gives the data set and experimental results, and discusses the structural channel set for extracting node features, neighborhood networks, and embedded feature information. Section 5 summarizes this article.

2. Related works

Key node identification is a key research direction in complex network analysis and has high practical value. In recent years, this issue has attracted wide attention from researchers in related fields. In the study of complex network analysis, the importance of nodes is usually called centrality. The greater the centrality value of the node, the higher the importance of the node in the network. According to the different methods, the existing research can be roughly divided into methods based on node influence and methods based on graph neural networks.

The methods based on node influence can be roughly divided into methods based on node local influence, global influence, and mixed node centrality influence. Among them, methods based on local influence, such as degree centrality (DC), H-index, neighbor classification, and other information multi-feature index methods, are simple and efficient, highly interpretable, stable to the disturbance of network structure, and can adapt to dynamic or incomplete network scenarios. However, based on local connection information, it will lead to a lack of global vision, and all neighbors are regarded as equally important, which cannot reflect the importance of neighbors themselves. Methods based on global influence, such as betweenness centrality (BC) and eigenvector centrality (EVC), can express the global structural role of nodes and predict potential key nodes more accurately [25-27]. However, they have high computational complexity, low efficiency, high sensitivity to the integrity of network structure, and insufficient robustness. The method of mixed node centrality is the clustering-enhanced Coreness algorithm [28]. This algorithm introduces the clustering coefficient of nodes on the basis of K-shell decomposition, which can take into account the topological position and local group structure of nodes, and the computational efficiency is also high. However, since the essence is still based on local topology, the description of global influence is insufficient, and it is easy to be affected by noise fluctuations to reduce robustness.

In recent years, graph neural networks (GNNs) have been widely used in key node recognition tasks, and influence prediction is achieved by learning the neighborhood representation of nodes [29-30]. InfGCN [31] can improve the quality of node representation by integrating multiple centrality indicators (such as degree centrality, K-shell, etc.) as the initial features of nodes to input the graph convolutional neural network model. Ou Y et al. [32] proposed to encode multi-level structural features into different channels, and input the propagation potential of learning nodes in the graph convolutional network, respectively, so as to improve the modeling ability of the multi-scale structure of the network. Some scholars used a convolutional neural network (CNN) to encode the neighborhood subgraph of nodes to extract local structural features and construct an end-to-end node influence regression model [33-35]. This method does not require manual design features and has strong automation capabilities. The above methods prove the effectiveness of the graph neural network applied to the key point recognition task [36-38]. However, there are still some shortcomings in the application of graph neural networks to key point recognition, such as: (1) GNN relies on the neighbor information of nodes for feature aggregation. When there are nodes missing in the graph structure, it will significantly affect the aggregation results, resulting in a decrease in recognition accuracy. (2) On large-scale networks, GNN model training and inference usually require a lot of computing resources, and computational efficiency is difficult to guarantee.

Inspired by the above related work, this paper focuses on how to improve the robustness, generalization ability, and computational efficiency of the key node identification method while ensuring the accuracy of the model identification by fusing the mixed centrality of nodes and graph neural network, to provide guidance solutions for efficient information dissemination, public opinion monitoring, and emergency intervention in the real world.

3. Methodology

In order to alleviate the problem of node representation deviation in the existing key node recognition algorithm, this paper proposes an improved algorithm NLGCN, combined with a graph convolutional neural network, on the basis of considering network heterogeneity. In this method, the concept of neighborhood set is introduced to strengthen the local structure representation, and the attention mechanism is introduced to distribute the weights of different channels, and the parameters are automatically adjusted to balance the local and global effects. The multi-source network information is integrated at a low cost, and the propagation effect of key nodes is improved. The model reconstructs the multi-scale centrality index and constructs the structural channel set. The local and global connectivity is jointly modeled by the graph convolutional network and the attention mechanism, and the channel weight distribution is automatically optimized, which effectively improves the ability to identify key nodes in different types of social networks. Figure 1 shows the basic framework of the NLGCN algorithm, which mainly involves three steps:

The first step is node feature extraction. In order to reduce the computational complexity of the representation of the structure of the node and ensure the fairness of each node in the representation process, NLGCN only uses the one-hop of the node neighborhood subgraph to generate the neighborhood matrix, and accommodates the feature information of nodes of different sizes through small local areas.

The second step is to embed the structural channel set of feature information. In order to construct the channel set of the NLGCN model training and prediction, the local and global influence indexes of nodes are integrated into the neighborhood network to create channels. Six channels are obtained by embedding node features through formulas, and then grouped into two channel sets based on local and global influences.

The third step is model prediction. Based on the model of GCN, this paper regards the influence ranking of key nodes in a social network as a regression problem, selects the BA scale-free network as the benchmark, uses SIR to generate labels, integrates local and global influence to construct features, and processes by self-attention mechanism and GCN layer (including a convolution layer).

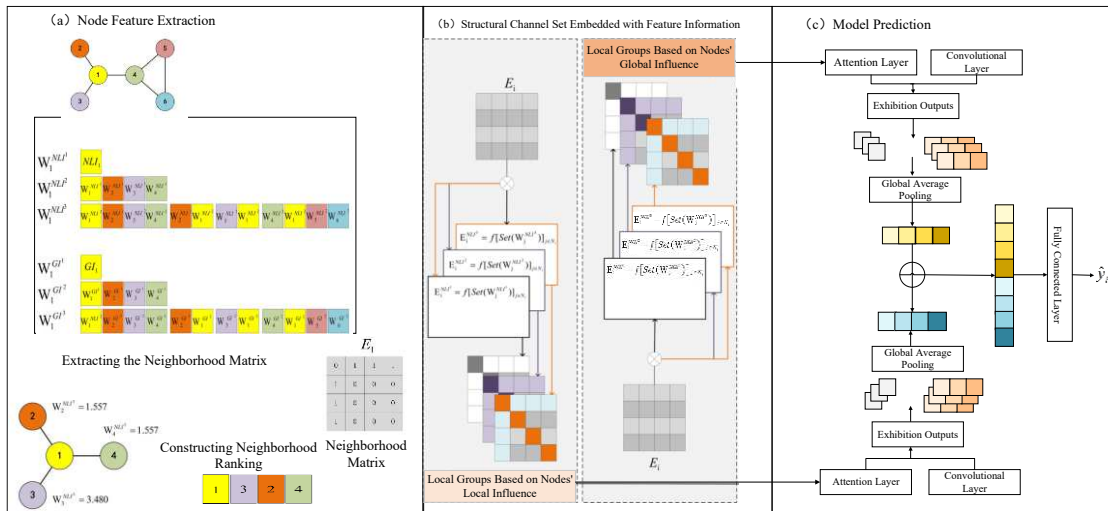


Figure 1: Basic Framework of NLGCN

3.1 Node feature extraction

In this paper, the K-order neighbor sum is introduced into the representation of the local influence of nodes, so as to further optimize the representation of the local structural features of nodes. At the same time, the node degree is used to replace the K-shell value in the GSM method to seek a lower time complexity. In addition, the algorithm also adds an adjustable parameter α with a value of $[0, 1]$ to the characterization of the global influence of nodes to control the moderate influence of the network, so as to effectively measure and screen the key nodes. The method of calculating local influence NLI_i (1), global influence NGI_i (2) and mixed influence $NLGI_i$ (3) is defined as:

$$NGI_i = \sum_{i \neq j} \frac{\sqrt{d(v_j) + \alpha}}{d_{ij}} \quad (1)$$

Where $d(v_j)$ is the degree of node j , d_{ij} is the shortest distance between node i and node j , α is a variable constant, according to the influence of the mobility control degree of the network on GI_i .

$$NLI_i = \frac{d(v_i) \times \lg \sum_{N=1}^K nei^N(v_i)}{n} \quad (2)$$

$$NLGI_i = NGI_i \times NLI_i \quad (3)$$

However, when the neighbor order of the node is larger and larger, its influence on the node is usually smaller and smaller. Therefore, the new index proposed above should have different weights when fitting the node influence function. Inspired by this, an attention mechanism is introduced in front of the convolutional layers of two independent units to enable the model to learn independently and pay more attention to the indicators that contribute more to the influence of nodes; that is, a larger weight is given in the final nonlinear function. Therefore, in different types and scales of social networks, the model can better learn the effective representation of the network to identify key nodes. The indicators based on local influence and the indicators based on node global influence are defined as follows:

$$W_i^{NLI^1} = \frac{d(v_i) \times \lg \sum_{N=1}^K nei^N(v_i)}{n} \quad (4)$$

$$W_i^{NLI^2} = W_i^{NLI^1} + \sum_{j \in N_i} W_j^{NLI^1} \quad (5)$$

$$W_i^{NLI^3} = W_i^{NLI^2} + \sum_{j \in N_i} W_j^{NLI^2} \quad (6)$$

The relationship between nodes and their neighbors is a key factor in information dissemination under any diffusion model. Therefore, firstly, three values of $W_i^{NLI^1}$, $W_i^{NLI^2}$ and $W_i^{NLI^3}$ are extracted for each node according to the local influence NLI of the node and the neighbor connection. Firstly, the sum of local influences $W_i^{NLI^1}$ is calculated, and then another centrality index is obtained according to the sum of local influences of neighbors, which enables the model to obtain additional information about its neighbors in the next hop neighbor. Finally, the $W_i^{NLI^2}$ index is used to provide additional information $W_i^{NLI^3}$ from the next hop neighbor, update the different hop information based on the neighborhood and enhance the strength evaluation of the node again.

$$W_i^{NGI^1} = \sum_{i \neq j} \frac{\sqrt{d(v_j) + \alpha}}{d_{ij}} \quad (7)$$

$$W_i^{NGI^2} = W_i^{GI^1} + \sum_{j \in N_i} W_j^{NGI^1} \quad (8)$$

$$W_i^{NGI^3} = W_i^{GI^2} + \sum_{j \in N_i} W_j^{NGI^2} \quad (9)$$

Similar to the previous step, the first centrality index $W_i^{NGI^1}$ based on the node global influence GI_i is defined. Then, to obtain the additional information of neighbors with different properties to achieve the consistency of adjacent subnets, another centrality index $W_i^{NGI^2}$ is obtained by summing the global influence of first-order neighbors. Finally, $w_i^{NGI^2}$ is generalized to $W_i^{NGI^3}$ to obtain its next update value to provide additional information from the two-hop neighbor subnet.

3.2 Extract neighborhood network

In order to reduce the computational complexity of the representation of the structure with the node and ensure the fairness of each node in the representation process, this paper uses the one-hop of the node neighborhood subgraph to generate the neighborhood matrix, and accommodates the node feature information of different scales through small local areas. This strategy avoids the rich club phenomenon generated by the BFS algorithm and alleviates the representation deviation of the node neighborhood matrix to a certain extent. In addition, if the neighborhood matrix only relies on the degree of the nodes to rearrange the nodes of the same hop, the simplicity of the index may make some nodes with the same degree lack discrimination, which will increase the randomness of node selection. Considering the interaction characteristics of nodes that should be paid attention to in social networks, Formula (6) is used instead of degree reweighting to distinguish all nodes.

Based on the above practice, for each node that needs to be measured, the model will filter out its one-hop neighbor and select the top L nodes according to the formula (6), the neighborhood matrix $(a_{ij})_{(L+1) \times (L+1)}$ is generated according to the node itself and the L one-hop neighbors. If there is an edge between the node i and the node j , let $a_{ij} = 1$, otherwise let $a_{ij} = 0$. If the total number of neighbors of some nodes is less than L, the matrix is expanded with 0 to ensure that each node has a neighborhood matrix of the same size.

3.3 Structure channel set embedded feature information

This paper will construct a channel set for the NLGCN model training and prediction, and create two channel sets by integrating the local influence index and global influence index of nodes into the neighborhood network generated by 3.2. The model will provide information directly related to the node based on the diameter of the channel set, and represent the information related to the node's neighbor in the remaining unit. Since the influence of direct interaction between neighbors and nodes is usually greater than the influence of interaction between neighbors on nodes, the following channels can be created based on the generated neighborhood matrix:

$$E_i^{(NLI^t)} = \begin{cases} W_i^{NLI^t}, & i = j \\ a_{1j} W_j^{NLI^t}, & i = 1, j = 2, 3, \dots, L+1 \\ a_{i1} W_i^{NLI^t}, & i = 2, 3, \dots, L+1, j = 1 \\ a_{ij}, & \text{otherwise} \end{cases}, i, j \in [1, L+1] \quad (10)$$

$$E_i^{(NGI^t)} = \begin{cases} W_i^{NGI^t}, & i = j \\ a_{1j} W_j^{NGI^t}, & i = 1, j = 2, 3, \dots, L+1 \\ a_{i1} W_i^{NGI^t}, & i = 2, 3, \dots, L+1, j = 1 \\ a_{ij}, & \text{otherwise} \end{cases}, i, j \in [1, L+1] \quad (11)$$

Where $t \in [1, 3]$.

Through the above formula, the extracted node features are successfully embedded into the neighborhood matrix of the node to create six channels, and then they are grouped into two channel sets : the local group $E_i^{(NLI)} = \{E_i^{(NLI^1)}, E_i^{(NLI^2)}, E_i^{(NLI^3)}\}$ based on the local influence of the node, the global group $E_i^{(NGI)} = \{E_i^{(NGI^1)}, E_i^{(NGI^2)}, E_i^{(NGI^3)}\}$ based on the global influence of nodes.

3.4 Model prediction

This paper uses a model based on a graph convolutional neural network (GCN), which regards the ranking problem of calculating the influence of key nodes in social networks as a regression problem. In order to ensure that the model can be used for most social networks as much as possible, this paper selects the synthetic network to obtain the input and label of the model. In the existing research, the BA scale-free network is widely considered to be close to the real-world network because of its power-law degree distribution characteristics. The model trained under this network can usually be smoothly extended to the real-world network and has good scalability. Therefore, a BA scale-free network with 8000 nodes and an average degree 4 is used as the benchmark network for model training. In addition, for each node, the size of its neighborhood matrix is limited to L, and the SIR model is used to generate real labels for it. Therefore, the input of the NLGCN model can be represented by a set of structural channels embedded with feature information. In the selection of feature information, the concepts of local influence and global influence are integrated, and the feature representation based on the two is constructed and divided into two sets with a capacity of 3. Before feeding to the convolutional layer, the model first introduces a self-attention mechanism. It first calculates the attention score through a fully connected layer, and then normalizes these scores to obtain the attention weight. Finally, the weight is applied to the feature tensor to obtain the weighted feature representation. After that, the model creates a GCN layer composed of a convolution layer, a batch normalization layer, an activation function layer, and a pooling layer, in which the attention mechanism is called for output.

4. Experiments

4.1 Dataset

To make a more accurate evaluation of the proposed NLGCN framework, the Networkx library function is used to generate a larger BA scale-free network than the model training, which is compared with the benchmark algorithm to confirm that the new indicators based on the local influence and global influence of nodes can complement each other with the graph convolutional neural network. For real-world social networks, choose five new social networks of different sizes: Facebook, Twitch, Gnutella, Email, and Wikipedia. For the synthesized BA scale-free network, a network with 8000 nodes and an average degree of 4 is generated. Table 1: Overview of Datasets and Their Statistical Characteristics summarizes the statistical properties of real-world social networks and synthetic networks. Here, $|M|$ denotes the number of edges in the network, $|N|$ denotes the number of nodes in the network, $\langle K \rangle$ represents the average degree of all nodes, K_{max} is the maximum node degree, and $\langle CC \rangle$ denotes the average clustering coefficient of the network. Meanwhile, the brief introduction to social networks is as follows: follows:

Table 1: Datasets and Their Statistical Features

Dataset	$ M $	$ N $	$\langle K \rangle$	K_{max}	$\langle CC \rangle$
Facebook	45266	5793	15.63	575	0.173
Twitch	37304	4385	17.01	1229	0.166
Gnutella	26013	8114	6.41	102	0.009
E-mail	5451	1133	9.165	71	0.220
Wikipedia	36101	2277	31.69	739	0.669
BA-8k	15996	8000	4.00	273	0.006

Facebook organization Networks: originated from an information technology company that provides products and services to customers around the world. The nodes in the network represent the Facebook accounts of employees in North America, South America, Asia, and Eastern Europe, and the edges represent the friendships between users.

Twitch Social Networks: This network is essentially a user network that broadcasts games live on the Twitch platform in a certain language. The node itself represents the user, and the edge represents the interaction between users, such as likes, attention, collection, and so on.

Gnutella P2P Networks: A series of Gnutella point-to-point file sharing networks over a certain period. Nodes represent hosts in the network topology, and edges refer to links between hosts that interact with each other.

E-mail Network: This data set is taken from an e-mail interaction between students and faculty in a university. It has 1133 nodes and 5451 edges. Each node corresponds to an individual in the university, one-to-one, and the connection between the nodes indicates that the two individuals have at least one email exchange.

Wikipedia Article Networks: From Wikipedia, it represents the page application network under a specific topic. The nodes in the network are regarded as articles, and the edges represent some kind of citation relationship between articles.

4.2 Analysis of parameters

The model training of this algorithm uses a BA scale-free network to train NLGCN, which is a network composed of 8000 nodes with an average degree of 4. At the same time, an infection probability $\beta = 1.5\beta_c$ is selected, and the SIR model is used to extract the node label and make it a training data set. In the training of NLGCN, the choice of the size L of the neighborhood matrix directly affects the performance of the model. Therefore, it is necessary to analyze the parameters of L to obtain a graph neural network model with relatively good performance. Inspired by this, the Kendall correlation coefficient value of the node ranking list generated by the NLGCN and SIR models is studied by using the BA scale-free network of the same size but with different node average degrees (this paper chooses 2,6,10). In the parameters analysis, the size of L is given to be 10 to 60, and the interval optimal parameters are determined based on the key node ranking list predicted by NLGCN and the results obtained by the SIR model under different infection probabilities. When L=40, the performance of the trained model is better than other L values. At the same time, as the node degree increases, the gap will become more obvious.

In addition, since NLGCN takes GCN as the core, the setting of Kernel size and Max-pooling directly affects the training process and training results of the model. In order to ensure that the selected parameters have strong interpretability and can be better extended to different types of social networks, we try to test different model results on the data set E-mail network, and the BA scale-free network with 4000 nodes and an average degree of 10. By selecting different Kernel sizes and Max-pooling sizes (this paper selects 2,4,6) to train the key node ranking list of NLGCN prediction and

the node ranking list of SIR model simulation under different infection rates, the average Kendall τ is used to evaluate the performance of the model. Figure 2 shows the test results under different NLGCN models. The results show that when the Max-pooling size is fixed at 2, the model performance of NLGCN still has stable and excellent performance under different Kernel sizes, and both of them achieve the optimal prediction results of the model under the current parameter interval when the Kernel size is 2. Correspondingly, if the Kernel size is first set to a fixed value, the prediction results continue to deteriorate as the Max-pooling size continues to increase. Based on the above model parameter analysis results, the Kernel size and Max-pooling size can be set to 2 when training the NLGCN model, so that it can ensure the best performance of the model as much as possible to cope with the next real social network data set.

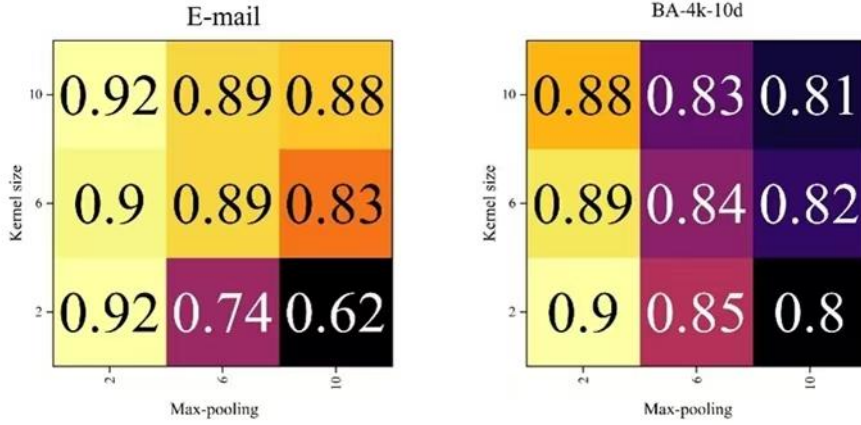


Figure 2: Analysis of the Impact of Kernel Size and Max-pooling on the Performance of the NLGCN Model

4.3 Comparison of key nodes

Based on the benchmark method, this paper uses the benchmark algorithm and the NLGCN algorithm to output the top 15 key nodes on six networks. By comparing the results, it is proven that the NLGCN algorithm is feasible in the identification of key nodes in social networks. According to Table 2, it can be seen intuitively that in the Facebook network, NLGCN has a high similarity with the Top-15 key nodes produced by the benchmark algorithm. For example, there are 13 node repetitions between NLGCN and EVC, 8 node repetitions between NLGCN and the LGC algorithm, and 7 node repetitions between NLGCN and the GIC algorithm. In the Twitch network, NLGCN, CNC, and LGC jointly identify 11 nodes as key nodes, and there are also 10 duplicate nodes compared with BNC and PRC. In the Gnutella network, there are 10 repeat nodes between NLGCN and EVC, and 9 repeat nodes between CNC, GIC, and NLGCN. Compared with LGC and PRC, the number of repeat nodes is 6, 5 and 4, respectively. In the E-mail network, there are 13 repeat nodes between NLGCN and EVC, 10 repeat nodes between NLGCN and GIC, 9 repeat nodes between NLGCN and LGC, PRC, PRC, and 6 repeat nodes between NLGCN and BNC, CNC. In the Wikipedia network, NLGCN still produces the same 10 key nodes as EVC. In the BA-8k scale-free network, the number of repetitions of the Top-15 key nodes of the NLGCN and the benchmark algorithm reaches the peak. Compared with the CNC and GIC algorithms, the NLGCN reaches 14 repeat nodes. Compared with EVC, the number of repetitions is 13. Compared with BNC, NLGCN has 13 repeat nodes, and even compared with the remaining methods, there are 10 repeat nodes.

Table 2: Summary of the Top-N Key Nodes Using Different Methods

Facebook							Twitch						
BNC	CNC	EVC	GIC	LGC	PRC	NLGCN	BNC	CNC	EVC	GIC	LGC	PRC	NLGCN
1486	2958	185	185	185	3539	185	1801	1801	3185	1801	1801	1801	1801
2958	1486	713	1135	989	1445	713	647	2426	3580	3185	647	647	2426
1926	18	1100	220	525	3403	1100	1726	647	908	1830	2426	2426	647
184	989	220	210	2958	3729	1103	913	1917	2120	1301	1726	1726	908
411	144	186	186	144	1700	780	599	908	234	3349	599	599	1830
3703	75	161	525	1100	1662	220	1277	3185	3744	1586	913	569	1917
1316	1863	525	161	713	3277	161	2426	913	1801	320	569	913	2274
185	2488	1103	1148	1135	1486	186	569	1726	2778	1457	1917	1277	320
75	1157	780	780	210	989	276	1917	599	887	1800	908	1917	1726
144	3976	489	276	220	1413	481	343	3744	1917	1181	1277	343	3185
1445	1408	693	481	186	2958	698	209	209	4278	2966	209	908	599
18	2968	481	1143	489	144	693	2615	3580	2536	3655	2615	209	913
1157	2209	276	171	538	1412	241	2511	320	4338	1300	343	2615	460
3278	2742	241	1134	161	3278	173	908	2120	1830	172	1457	2511	2615
989	2961	171	619	780	411	489	3185	2615	1780	2965	2274	1457	1457
Gnutella							E-mail						
BNC	CNC	EVC	GIC	LGC	PRC	NLGCN	BNC	CNC	EVC	GIC	LGC	PRC	NLGCN
4544	100	273	276	342	7434	80	333	333	105	105	105	105	105
3742	342	80	342	276	5971	110	105	23	16	333	333	333	16
175	101	342	101	273	39	273	23	105	196	42	42	23	196
3760	271	276	273	39	192	341	578	42	204	23	23	42	42
2141	36	110	110	192	81	101	76	41	42	76	41	16	3
5971	276	101	39	36	36	35	233	76	49	41	16	41	23
39	110	341	36	80	342	34	135	233	56	233	196	196	333
3611	34	34	34	81	273	113	41	52	116	52	233	233	204
164	273	39	321	271	276	342	355	135	333	196	76	76	49
100	39	321	35	164	349	189	42	378	3	3	21	21	332
7434	164	35	82	101	164	41	378	3	128	332	135	24	41
36	35	187	102	349	100	102	429	54	23	135	355	135	21
33	80	271	187	100	271	320	52	137	21	299	49	354	116
82	341	36	349	187	80	276	564	39	206	49	354	355	56
272	187	274	81	82	33	321	196	134	210	128	24	49	128
Wikipedia							BA-8k						
BNC	CNC	EVC	GIC	LGC	PRC	NLGCN	BNC	CNC	EVC	GIC	LGC	PRC	NLGCN
1940	1940	1977	2032	1977	1977	221	6	6	6	6	6	6	6
1977	1977	1940	2160	1940	1940	2031	3	3	3	3	3	3	5
1742	1742	1742	1986	1742	1742	1715	5	5	5	5	8	8	3
2250	1863	2264	1293	2264	2264	1913	8	4	4	8	5	5	4
1912	2250	2247	1642	2247	2247	2059	1	12	12	12	12	1	12
2247	2247	1357	1685	221	1357	1977	12	8	16	4	16	16	8
1709	2264	221	951	1357	2250	1940	16	1	11	16	1	12	11
1847	2165	2250	1068	1715	1334	2178	4	11	8	11	32	32	27

1863	1357	1715	2222	2250	1861	1188	11	16	135	27	11	87	7
1924	653	1861	1468	1924	653	891	68	27	7	68	4	51	16
1597	777	1924	312	1861	2111	2180	34	7	27	7	87	11	135
1715	1861	1334	907	1334	2235	532	15	135	1	2	51	34	1
2264	1464	2031	2181	2031	1912	2247	32	2	68	135	68	68	2
868	68	1913	278	1913	221	2264	51	68	9	51	20	4	19
1202	2111	2059	353	2059	675	1742	27	31	45	19	15	20	68

Based on the above experimental results, it can be found that the NLGCN algorithm has found key nodes that are partially repeated with the benchmark algorithm in five real social networks and the synthesized BA scale-free networks with social network characteristics, which, to some extent, demonstrates that the method can identify key nodes. In addition, it should be noted that NLGCN produces Top-15 results that are completely different from some benchmark algorithms in some cases, indicating that NLGCN can further distinguish key nodes in social networks by considering the local and global effects of nodes and neighborhoods.

4.4 Performance test based on the SIR model

In order to enable NLGCN to discover the most critical nodes in the social network, like most algorithms, it is also possible to evaluate the influence of each node more accurately to make up for the shortcomings of some traditional algorithms in terms of discrimination. Here, the SIR model is still used to evaluate the propagation efficiency of nodes in social networks. Based on the mean field theory, the SIR lists compared here are all generated under $\beta = 1.5\beta_c$, and due to the large scale of the network, the parameter $K = 3$ is specified when calculating the global influence GI_i of the network, that is, the sum of the third-order neighbors of the nodes. Through this comparison, the similarity between the proposed algorithm and the real propagation process of nodes can be tested. If the algorithm achieves similar performance with the SIR model under less time consumption, the performance of the algorithm is worthy of recognition. Based on the above basis, the recovery probability $\mu = 1$ is still set, and the following two experiments are carried out:

In the first experiment, BNC, CNC, EVC, GIC, LGC, PRC, and NLGCN are used to calculate the influence of all nodes in each network. After ranking the nodes in reverse order, each node is input into the SIR model as a seed node to simulate the propagation process. Let the time step $t = 1000$, and perform 500 independent experiments on each node to obtain the average number of nodes it can infect. There is no doubt that the number of infected nodes is closely related to the real influence of nodes. Therefore, if the node list produced by the algorithm is closer to the real simulation results, the infection curve of the output will show a decreasing characteristic. The infection curve of each method in each social network is shown in Figure 3.

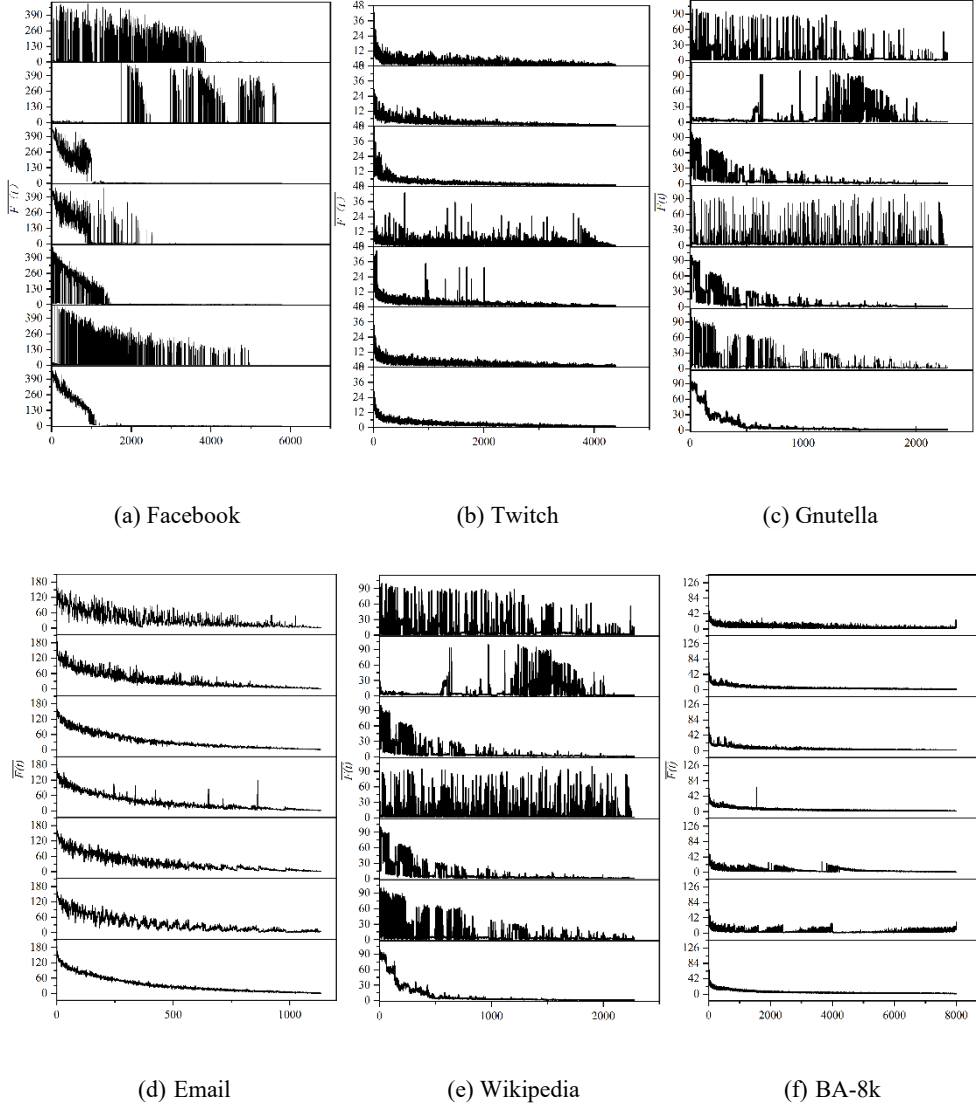
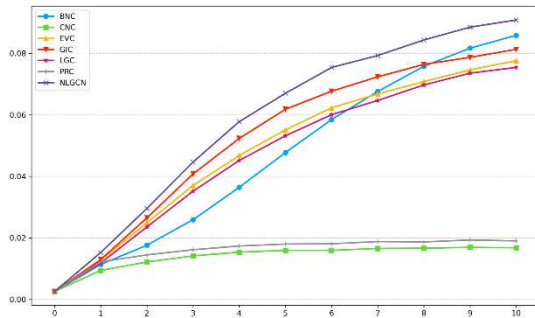


Figure 3: Node Propagation Impact under NLGCN and Benchmark Algorithms

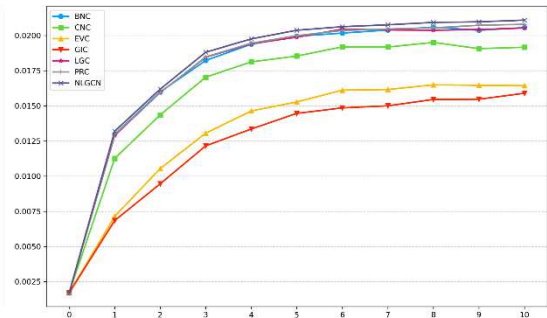
It can be intuitively found that in Facebook, Gnutella, and Wikipedia networks, the curve generated by the benchmark method is very unstable. In most cases, they neither accurately identify the key nodes in the social network, nor better distinguish all nodes. NLGCN predicts key nodes and outputs the final infection curve, which greatly improves the above phenomenon. Although there are still some fluctuations in the image (such as Facebook and Wikipedia), it is greatly converged compared with the benchmark method, and the three curves show a steady decreasing trend. In addition, in the networks of Twitch, Email, and BA-8k, each method has achieved good results. They no longer have the situation of great fluctuation and fuzzy trend in the previous network, but after careful observation, it can be found that NLGCN still shows its advantages. For example, in the Twitch network, NLGCN alleviates the 'pin insertion' phenomenon of GIC and LGC algorithms, and the degree of fluctuation is further reduced compared with PRC and NLGC algorithms. This phenomenon is more obvious in Email and BA-8k networks. It can be seen that the infection curve produced by NLGCN is very close to a decreasing function at this time. This shows that in these two networks, NLGCN has reached the effect of the simulation model similar to SIR, but the time consumption of the former is greatly reduced compared with the latter. The model trained in a network has achieved good performance in social networks with different scales and different

properties, and the universality and robustness of NLGCN have been preliminarily reflected.

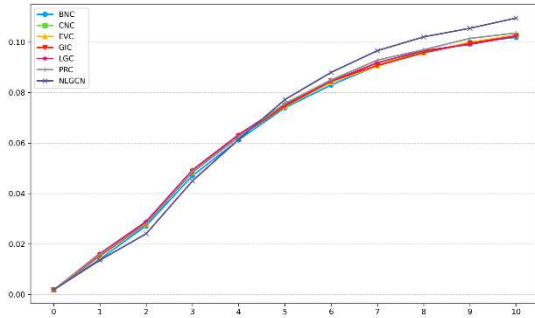
In the second experiment, the Top-N node groups produced by various methods are still input into the SIR model at the same time, and the time step $t = 10$ is fixed to calculate the propagation ability of Top-N nodes, and then the authenticity of these key nodes is demonstrated. Based on the above ideas, taking into account the Top-15 key nodes produced by each method listed in Table 2, these 15 nodes are added to the SIR model of each network as a seed node group. The difference in infection probability will directly affect the characteristics of the propagation model. If the selected key nodes can infect as many other nodes as possible under different infection conditions, these key nodes can be considered effective. To avoid the contingency of the experiment, the infection probability $\beta = \gamma \times \beta_c, \gamma \in [1, 1.9]$ is changed to obtain the average infection node of the node group under each step. The specific experimental results are shown in Figure 4.



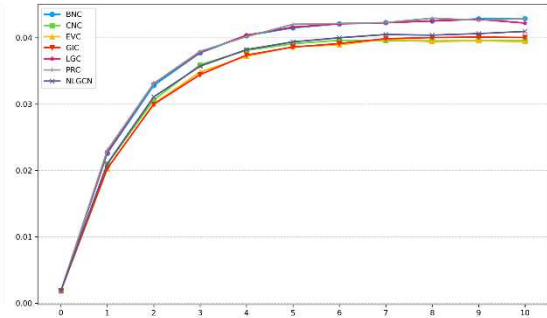
(a) Facebook



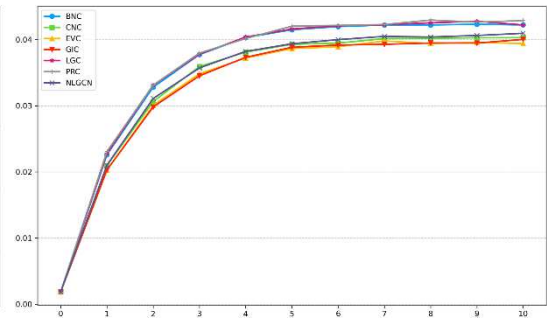
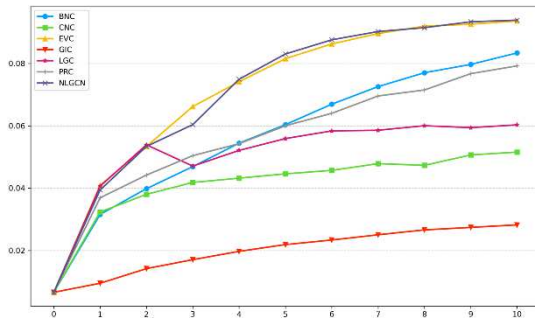
(b) Twitch



(c) Gnutella



(d) Email



(c) Wikipedia

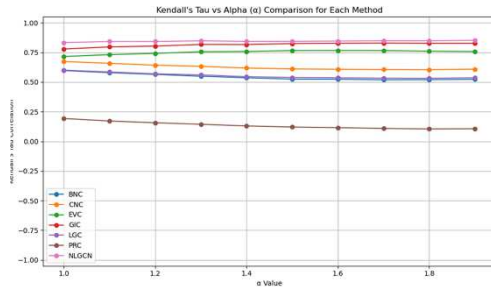
(f) BA-8k

Figure 4: Propagation Impact of Top-N Nodes by Various Methods with Fixed Step Size

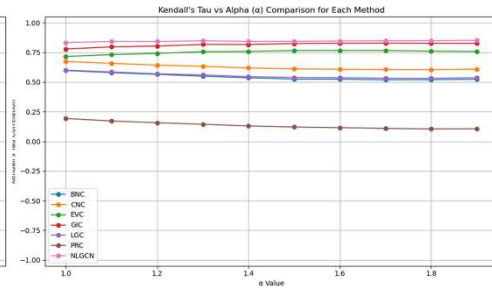
Based on Figure 4, it can be found that the Top-15 key nodes produced by the NLGCN method show excellent propagation performance in their respective social networks, and the performance is further improved compared with the NLGC algorithm. Especially in the Gnutella network, although the Top-15 key nodes screened by NLGCN are not ideal at the beginning of the propagation efficiency, with the gradual increase of the step size, the infection scale expands rapidly and exceeds all the benchmark algorithms, which shows that it can find the 'bridge' nodes in the network. In summary, by comparing and introducing nodes into the SIR model, it is proven that the key nodes screened by NLGCN are more efficient than the benchmark algorithm.

4.5 Performance comparison based on the Kendall correlation coefficient

Similar to the previous section, this section also uses the Kendall coefficient τ to compare the accuracy of NLGCN with other benchmark methods in five real-world social networks and one synthetic network. Figure 5 shows the performance analysis of the ranking list generated by various algorithms and the ranking list generated by the SIR model. It can be seen that under different network conditions, NLGCN is more efficient than the benchmark algorithm most of the time, which shows that the node ranking produced by NLGCN is closer to the node ranking produced by SIR. In addition, a significant advantage of NLGCN is more specifically reflected here, that is, it performs well in data sets with different average node degrees, network size, and the BA-8k network used for training. In other words, the performance of NLGCN is less affected by changes in network size and average node degree. Relatively speaking, the GIC algorithm performs well in Facebook, Email, BA-8k network, but not in Twitch, Gnutella, and Wikipedia. Similar to the CNC algorithm, it has achieved good performance in Twitch, Gnutella, Email, BA-8k network, but in Facebook, the Wikipedia network performance is flat. This phenomenon also shows that some methods can achieve better performance in some networks but are not universal, while the NLGCN model can still obtain more accurate results by using training sets that are different from the test set structure, which means that NLGCN has the potential to migrate to other fields compared with previous algorithms.



(a) Facebook



(b) Twitch

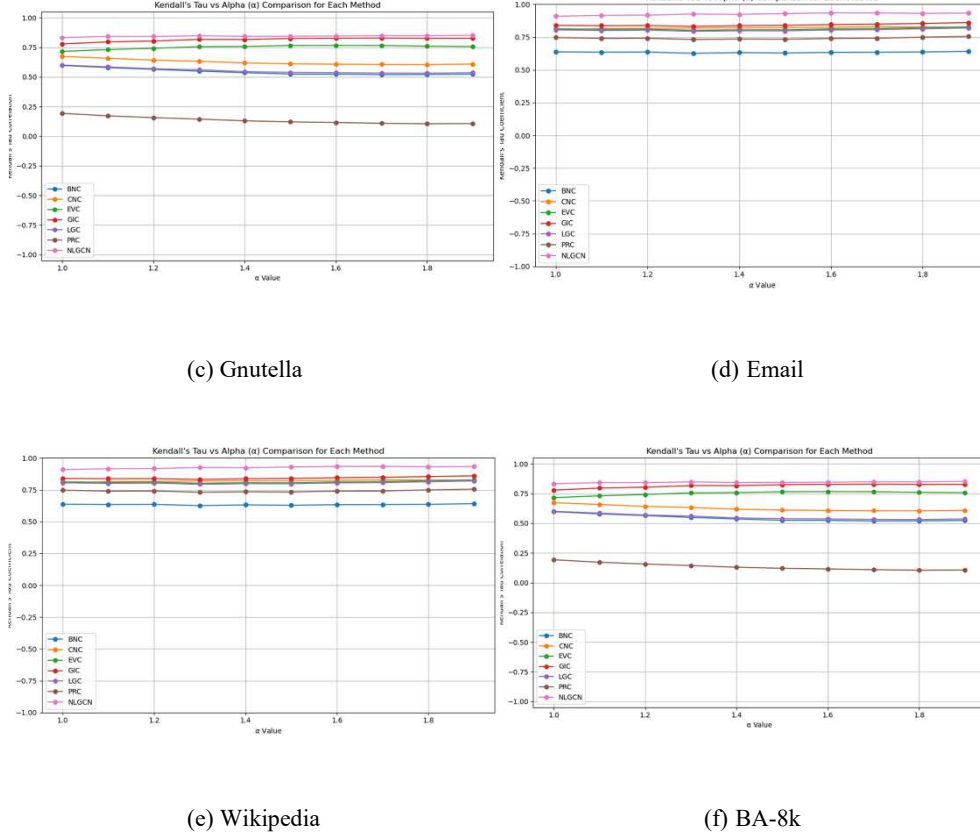
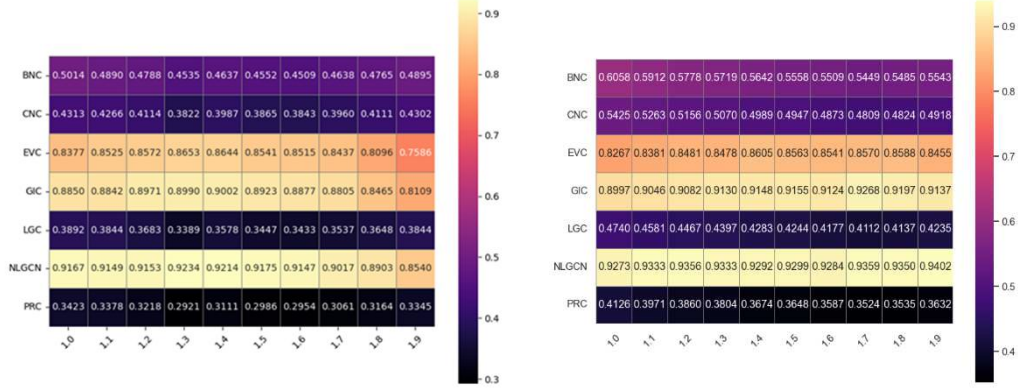


Figure 5: Kendall Correlation Coefficients between Node Rankings Generated by Various Methods and the SIR Mode

4.6 Performance comparison based on Avg_{MAP} metric

Similarly, this experiment will select the top 10 % and 20 % nodes sorted by the nodes given by each algorithm for different networks, and compare them with the top 10 % and 20 % of the SIR list under different propagation parameters to determine whether these nodes appear and correspond to the label position one by one. Figure 6 (a) shows all the Avg_{MAP} results that predict the top 10 % nodes in real social networks and synthetic networks. It can be seen that NLGCN ranks first in these six networks and is superior to other benchmark methods. It is worth noting that NLGCN, based on a deep learning model, achieves more than 80 % stability in these six networks, but the performance of benchmark algorithms based on network topology alone is extremely unstable. At the same time, EVC, GIC, and other algorithms perform well in the synthetic network, while LGC and other algorithms perform well in the social network. In contrast, it shows that NLGCN does not have too many requirements for network characteristics like the former. To further explore the performance of the method, Figure 6 (b) shows the Avg_{MAP} results of the top 20 % nodes predicted by the same method, where NLGCN still maintains better performance than other benchmark algorithms, which also confirms the stability and effectiveness of the method. At the same time, it can be observed that the performance of LGC, EVC, BNC, and other algorithms is greatly improved with the increase of the sampling node ratio, which also shows that these algorithms have the problem of weak ability to predict the top key nodes, while NLGCN avoids this phenomenon.



(a) $AvgMAP@10\%$

(b) $AvgMAP@20\%$

Figure 6: Performance Analysis of Identifying the Top 10% and 20% Key Nodes Based on $AvgMAP$

4.7 Robustness analysis

Considering the network robustness index, the nodes are removed in turn in the corresponding network according to the node ranking given by each algorithm, and the number of nodes in the largest connected subgraph of the current network is selected as the robustness index, aiming to evaluate the ability of the algorithm to find those 'bridge' nodes in the social network. Similar to the previous practice, Figure 7 shows the relationship between the number of nodes and the number of nodes in the largest connected set of the network after the six networks of different sizes and different properties are arranged in descending order of each algorithm. As shown in Figure 7, the BNC, CNC, and PRC indices in the Facebook network caused a rapid decline in the function value after removing a small number of nodes. Although the performance of NLGCN was not as good as the former at the beginning of the experiment, it still had obvious advantages compared with EVC, GIC, LGC, and other algorithms. At the same time, it is worth noting that as time goes on, the function corresponding to NLGCN is closest to the value of 0, which also reflects that the NLGCN method is better than other benchmark methods. As shown in Figure 7 (b) and Figure 7 (c), the PRC algorithm shows excellent performance here, which also benefits from its idea of considering network mobility to the greatest extent. However, at the same time, the NLGCN model achieves the performance of benchmark algorithms other than PRC, and the gap with PRC is very close. As shown in Figure 7 (d) and Figure 7 (f), in the Email and BA-8k networks, the results obtained by NLGCN are better, and the curve has been basically consistent with the curve of the optimal benchmark algorithm. In the final Wikipedia network shown in Figure 7 (e), NLGCN maintains the most significant impact on network robustness from the beginning of the experiment to the end of the experiment, which is superior to other benchmark algorithms.

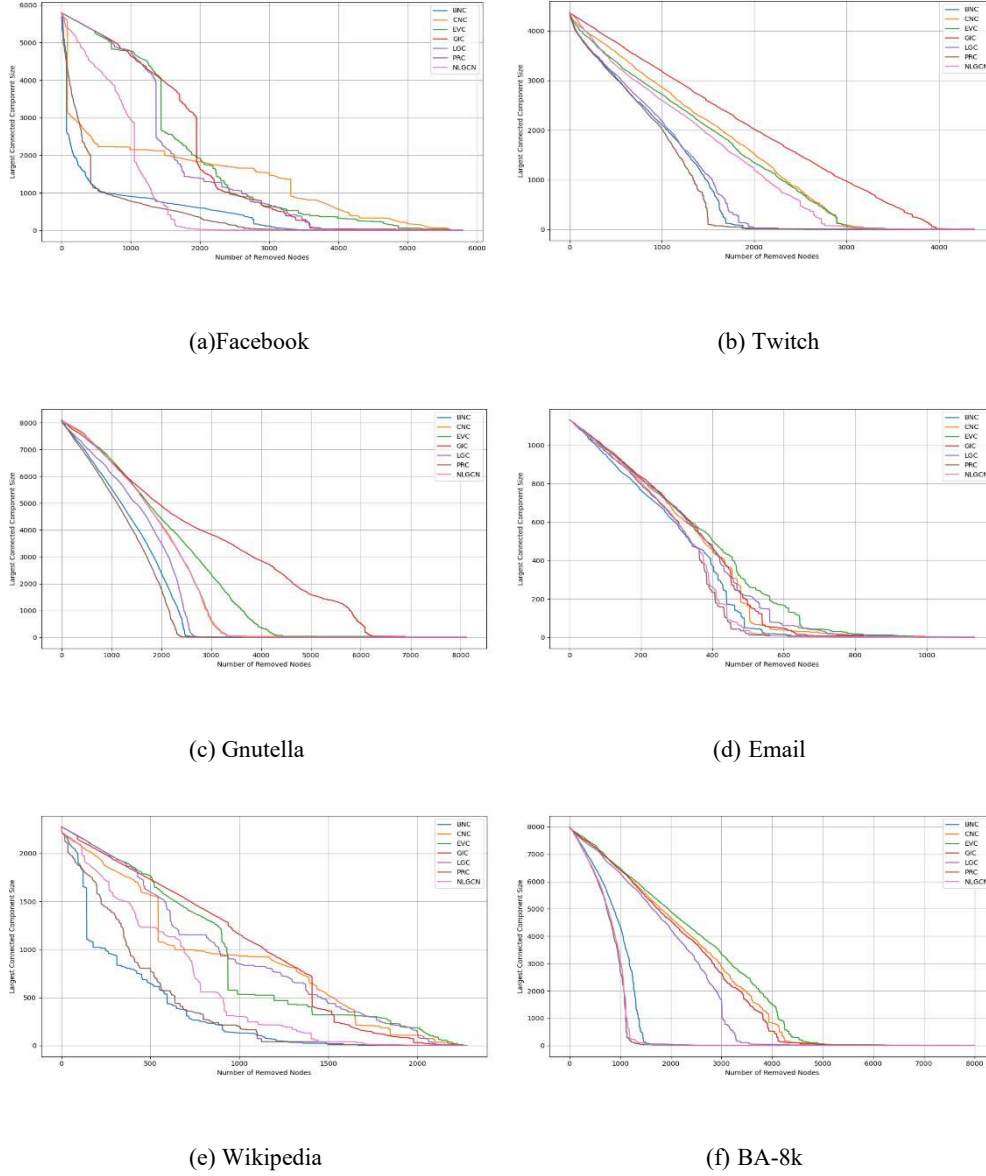


Figure 7: Changes in Network Robustness of Various Social Networks After Node Removal

By analyzing the above experimental results on synthetic networks and multiple real networks, it can be seen that the key nodes identified by the NLGCN method have high authenticity, and it can better screen out the 'bridge' nodes connecting each individual in the network, which is closely related to the idea of integrating the global influence of nodes. At the same time, it can be observed that the performance of BNC and GIC algorithms in different networks fluctuates greatly, and the time consumption of some algorithms is too high in the actual operation process. NLGCN also avoids these phenomena well, so it can identify key nodes in social networks more quickly, stably, and accurately than other benchmark indicators.

5. Conclusion

In this paper, a key node recognition method based on a graph convolutional neural network is proposed to achieve both computational efficiency and recognition accuracy in complex networks. Firstly, the local influence NLI and global influence GI of nodes are used to redefine the multi-

scale centrality index to complete the extraction of node feature information. Secondly, the neighborhood matrix based on the first-order neighbor is used to accommodate the node feature information, which avoids the defect of the neighborhood matrix uncertainty in the BFS algorithm, and also captures the neighborhood connectivity of the node interaction as much as possible, thus completing the construction of the structural channel set embedded with the feature information. Finally, when the node information is transmitted to the graph convolutional neural network model for regression prediction through the above structural channel set, the attention mechanism is introduced to better solve the problem of manually assigning parameters weights. After training with a BA scale-free network with social network characteristics, relevant experiments are carried out. The experimental results show that the proposed algorithm has the ability to identify key nodes in social networks of different sizes and types. In future work, we hope to extend NLGCN to more complex heterogeneous networks by introducing a graph attention mechanism to encode node types.

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Data Availability Statement

Data will be provided upon request.

Declarations

Conflict of interest: The authors declare that they have no conflict of interest.

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